DATA ENGINEERING

Data Catalogs Visual Guide

Find the right dataset in minutes, not Slack threads.

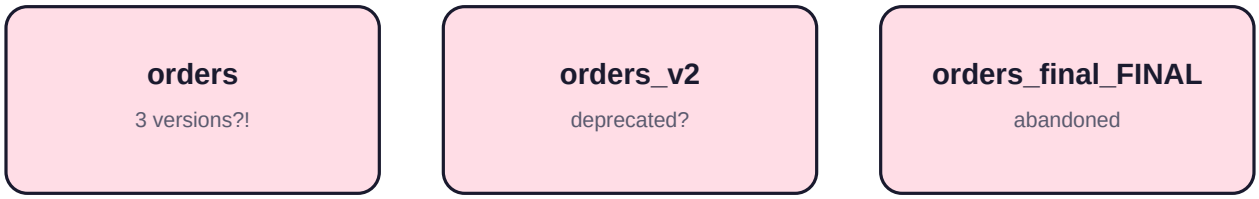
ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

The Discovery Problem

The average analyst spends a day a week just finding data.



Without a catalog: duplicate tables, tribal knowledge, and the same 'which orders table?' question every onboarding.

STEP 2/6

Catalog Anatomy

What a catalog records about every dataset.

Technical metadata

Schema, types, partitions, storage location — harvested automatically

Business context

Owner, description, glossary terms, SLAs — curated by humans

Usage signals

Query counts, popular columns, downstream dashboards — computed

Quality signals

Freshness, test status, incident history — from monitors

Automation harvests the technical layer; humans add the meaning. A catalog with only auto-harvested metadata is a phone book with no names.

STEP 3/6

Search That Actually Works

Analysts search like humans — the catalog must rank like one.

"monthly revenue by region"

natural-language search

Ranking signals: name match < description match < glossary term < certified badge < usage popularity. The most-queried, certified table should win — not the alphabetically first duplicate.

Certification badges ('verified by data team') are the strongest trust signal. Uncertified tables get a warning, not a ban.

STEP 4/6

Business Glossary

One definition per term, agreed once, referenced everywhere.

"Active customer"

purchased in last 12 months

"Revenue"

recognized, net of refunds, USD

Terms link to the columns and tables that implement them.

Two teams, one definition of 'revenue' — the end of dueling dashboards in the exec meeting.

The glossary is a social contract first, a feature second. Get finance and product to sign the definitions before encoding them.

STEP 5/6

Popularity Signals

Usage data tells you what's trusted — and what's dead.

Hot
queried daily, many users

Cooling
usage declining

Dead
zero queries, 90 days

Popularity drives deprecation: announce, migrate consumers, then drop. It also surfaces the 'golden' tables new hires should start from.

- Track: distinct users, query volume, downstream dashboard count
- Dead tables still cost storage and confuse search — deprecate them
- Lineage + popularity: deprecating a table shows every affected dashboard

STEP 6/6

Making the Catalog Stick

A catalog nobody opens is shelfware.

- Embed in the workflow: catalog links in BI tools, SQL editors, Slack
- Make owners curate: descriptions and certification are part of 'done'
- Seed with the top 50 tables — coverage of everything can wait
- Measure: search success rate, time-to-first-query for new hires
- Treat the catalog as a product with a roadmap, not a project

Start where the pain is: the 20 tables everyone asks about. A thin catalog people use beats a complete one they ignore.

STEP ✓I✓

Key Takeaways

Data Catalogs — Remember

- Catalogs solve discovery: one searchable place for technical metadata, business context, and trust signals.
- Automate the technical harvest; humans curate descriptions, owners, and glossary terms.
- Rank search by meaning and popularity, and badge certified tables as the trusted choice.
- A shared business glossary ends definition wars — get sign-off before encoding.
- Popularity signals drive deprecation; embed the catalog where analysts already work.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev